

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-09	Team No-04	Project Title- WarehouseIQ : Smart Warehouse & Inventory Management System
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1. Abstract

WarehouseIQ: Smart Warehouse&Inventory Management System is a database-driven system designed to efficiently manage warehouse operations and inventory. Traditional warehouse management can involve manual record keeping, which may lead to stock errors, difficulty in tracking products, and delays in updating inventory information.

The proposed system provides a centralized platform to manage products, suppliers, customers, stock levels, purchases, sales, and warehouse transactions. It maintains organized records in a relational database and allows users to add, update, delete, and retrieve information efficiently. The system can also monitor available stock, identify low-stock products, and maintain transaction history.

The project focuses on database design and backend development using tables, relationships, SQL queries, constraints, and database operations. By automating important inventory-related tasks, WarehouseIQ aims to reduce manual errors, improve data consistency, and make warehouse information easier to access and manage. The system can be extended in the future with reporting, analytics, notifications, and other smart inventory features.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030552	P.Gnaneshwar Reddy	
2520030401	Yashas	
2520030404	Venkatesh	

3. Frameworks / Platforms / Software Used

MySQL – Database Management System

SQL – Database queries and operations

Git/GitHub – Version control

Frontend – React + CSS

Guide Name and Signature (remarks if any) _____